

Core Java Master Interview Roadmap

Comprehensive Combined Topic Guide & Curriculum Breakdown

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Module 1: Java Execution & Architecture Mechanics

1. What is Java? & History of Java
2. Features of Java & Platform Independence
3. JDK, JRE, and JVM Architectural Components
4. How a Java Program Executes (Compilation vs. Execution)
5. Bytecode Mechanics & javap Disassembler
6. JIT Compiler, AOT Compilation, and Interpreter
7. JVM Internal Memory Architecture (Stack, Heap, Metaspace, PC Registers)
8. Memory Errors: StackOverflowError vs. OutOfMemoryError
9. Java Keywords & Identifiers Naming Conventions

 **Module 1 Mock Interview Included**

Module 2: Data, Variables, and Fundamentals

1. What is a Variable? (Declaration, Initialization, Scope)
2. Types of Variables (Local, Instance, and Static Variables)
3. Primitive Data Types (Integral, Floating-Point, Character, Boolean)
4. Memory Allocation & Ranges of Primitive Data Types
5. Default Values of Instance Variables vs. Uninitialized Local Variables
6. Operators in Java (Arithmetic, Relational, Logical, Bitwise, Shift, Ternary)
7. Operator Precedence & Associativity
8. Type Casting & Type Promotion Rules (Implicit vs. Explicit)
9. Input/Output Handling (Scanner, BufferedReader)
10. Control Flow Statements (Selection, Iteration, Jump)
11. Command Line Arguments (String[] args) **+ADDED**
12. Math & System Utility Classes **+ADDED**

 **Module 2 Mock Interview Included**

Module 3: Methods, Memory & Execution Mechanics

1. What is a Method? (Declaration, Signature, Return Types)
2. Stack Frame Push/Pop Operations during Method Calls
3. Java Pass-by-Value Execution Mechanics (Primitives vs Object References)
4. Method Overloading & Ambiguity Resolution Rules
5. Static Methods, Static Blocks, and Static Variables Execution Order
6. Constructors (Default, Parameterized, Copy Constructors)
7. Constructor Chaining & Overloading
8. The this Keyword (6 Major Use Cases)
9. Packages, Importing, Package Access Rules & Static Imports **+ADDED**
10. Access Modifiers (private, default, protected, public)
11. Enums (Enumerations with Constructors & Methods) **+ADDED**
12. Variable Arguments (Varargs: DataType... args) **+ADDED**

 **Module 3 Mock Interview Included**

Module 4: Object-Oriented Programming (OOPs) & Inner Classes

1. Classes & Objects (State, Behavior, Memory Layout)
2. Encapsulation & Data Hiding
3. Inheritance (Single, Multilevel, Hierarchical)
4. The `super` Keyword (Accessing Parent Fields/Methods)
5. Method Overriding & Dynamic Method Dispatch
6. Covariant Return Types in Overriding
7. Abstraction: Abstract Classes vs Abstract Methods
8. Interfaces (Default, Static, and Private Methods)
9. Object Relationships: Association, Aggregation, Composition
10. The `final` Keyword (Variables, Methods, Classes)
11. The Object Class Mechanics (`equals`, `hashCode`, `toString`)
12. The `equals()` and `hashCode()` Contract
13. Custom Immutable Class Creation
14. Inner Classes (Member, Static Nested, Local, and Anonymous Inner Classes) **+ADDED**

 **Module 4 Mock Interview Included**

Module 5: Arrays & String Handling Utilities

1. Arrays in Java (1D, 2D, Jagged Arrays) & Memory Allocation
2. Array Utility Methods (`java.util.Arrays`)
3. String Immutability & String Constant Pool (SCP) Mechanics
4. Literal Creation vs `new String()` Allocation
5. The `.intern()` Method Deep-Dive
6. String vs `StringBuilder` vs `StringBuffer`
7. Wrapper Classes, Autoboxing, and Unboxing
8. Integer Cache Mechanics (-128 to 127) & Null Pointer Risks
9. `StringTokenizer` & `StringJoiner` Utilities **+ADDED**

 **Module 5 Mock Interview Included**

Module 6: Exception Handling Mechanics

1. Exception Hierarchy (`Throwable`, `Exception`, `Error`)
2. Checked vs. Unchecked (Runtime) Exceptions
3. `try-catch` Blocks & Multi-Catch Syntax
4. Nested `try` Blocks & Stack Exception Propagation **+ADDED**
5. `finally` Block Execution Rules & Edge Cases
6. `throw` vs. `throws` Keywords
7. Creating Custom Checked and Unchecked Exceptions
8. `try-with-resources` & `AutoCloseable` Interface

 **Module 6 Mock Interview Included**

Module 7: Collections Framework & Cursors

1. Collection Hierarchy Overview (Collection vs Collections)
2. Collection Cursors: Iterable, Iterator, ListIterator, Enumeration **+ADDED**
3. List Interface: ArrayList vs LinkedList vs Vector
4. Set Interface: HashSet, LinkedHashSet, TreeSet
5. Queue & Deque Interfaces: PriorityQueue, ArrayDeque
6. Map Hierarchy: HashMap, LinkedHashMap, TreeMap
7. HashMap Under the Hood (Buckets, Indexing, Collision Resolution)
8. Fail-Fast vs. Fail-Safe Iterators (ConcurrentModificationException)
9. Sorting Elements: Comparable vs Comparator
10. Collections Utility Class Methods **+ADDED**

 **Module 7 Mock Interview Included**

Module 8: Multithreading & Concurrency Mechanics

1. Threads vs Processes & Thread Lifecycle States
2. Creating Threads (Thread Class vs Runnable Interface)
3. Thread Methods (start, run, sleep, join, yield)
4. Thread Priorities, Daemon Threads, and ThreadGroup **+ADDED**
5. Race Conditions & Synchronization Mechanics
6. Inter-Thread Communication (wait, notify, notifyAll)
7. Deadlocks (Creation, Detection, and Prevention)
8. The volatile Keyword & Atomic Variables
9. Concurrent Collections: ConcurrentHashMap Mechanics
10. Executor Framework, Thread Pools, Callable, Future, CompletableFuture

 **Module 8 Mock Interview Included**

Module 9: Java 8+ Modern Features & Date-Time API

1. Lambda Expressions & Functional Interfaces
2. Built-In Functional Interfaces (Predicate, Function, Consumer, Supplier)
3. Method References (Class::method)
4. Stream API Mechanics (Source, Intermediate, Terminal Operations)
5. Stream Operations (map, flatMap, 0, filter)
6. Optional Class for Null Safety
7. Date & Time API (LocalDate, LocalTime, LocalDateTime) **+ADDED**

 **Module 9 Mock Interview Included**

Module 10: Advanced Core Mechanics, GC & Annotations

1. Garbage Collection Algorithms & Heap Generations
2. Object Reachability & Reference Types (Strong, Soft, Weak, Phantom)
3. Java Serialization (Serializable, transient, serialVersionUID)
4. Marker Interfaces (Cloneable, Serializable, RandomAccess) **+ADDED**
5. Java Custom Annotations (@Retention, @Target) **+ADDED**
6. Reflection API & Native Methods Overview **+ADDED**

 **Module 10 Comprehensive Interview Included**

Core Java Interview Readiness Matrix

Topic Area	Primary Focus	Key Interview Focus Question
Module 2 (Variables)	Memory scope, primitives vs objects, precedence	"Why can't local variables have access modifiers?"
Module 4 (OOPs)	Inheritance rules, memory layouts, Object contracts	"Why override hashCode () when overriding equals () ?"
Module 7 (Collections)	Internal data structures, time complexity	"Walk me through how HashMap handles hash collisions."
Module 8 (Concurrency)	Locks, thread synchronization, thread pools	"What is the difference between volatile and synchronized?"